

Institute of Actuaries of India

Subject ST2 – Life Insurance

October 2015 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**(i) Surrender values should:**

- take into account policyholders' reasonable expectations
- treat both surrendering and continuing policyholders equitably
- at early durations, not appear too low compared with premiums paid, taking into account any projections given at new business stage
- should run into the maturity value
- take account of surrender values offered by competitors (and possibly also auction values, where available)
- not be subject to frequent change, unless dictated by financial conditions
- not be subject to significant discontinuities by duration
- not be excessively complicated to calculate, taking into account the computing power available
- be capable of being documented clearly
- avoid selection against the insurer

[4]**(ii)**

- The interest rate for discounting the future cash flows would be most affected given the economic outlook has changed.
- The discounting rate should be consistent with the current interest rates
- The extent of change would however depend on the future outlook as well since ideally surrender value should not be subject to frequent change
- The discount rate which would have been set at the time of product launch might already have taken into account change in future outlook, which would limit the change required in the discount rate
- Similarly, the future expense inflation under the prospective calculation also needs to be consistent with the interest rate
- If the interest rates have increased, customers may choose to surrender and invest in other higher yielding instruments, leading to loss of future expected profit. The basis should capture to allow for future expected profits recovery, at least part of it.
- The surrender expenses under the prospective valuation also need to be considered in view of the above.

[3]**[7 Marks]**

Solution 2:

i. Products may be written on the following main bases:

- conventional without-profits –
 - oldest form of writing insurance contracts,
 - characterized by fully guaranteed benefits,
 - usually, level regular premiums [except for immediate annuities, which would generally be paid for by single premium]
- with-profits –
 - where the policyholder has an entitlement to part or all of any future surplus which arises under the contract [or more widely to share in surplus arising within the with-profits fund].
 - The policy may be “conventional with-profits” or “accumulating with-profits”. The difference between the two is based on how bonuses are added to the policies
- unit-linked –
 - where the benefits are linked directly to the investment performance of a specified fund,
 - characterized by a lower level of guarantees on benefits and premiums
- Index-linked –
 - where the benefits are linked directly to a specified investment index or economic index, and
 - are guaranteed to move in line with the performance of that index.

[4]

ii.

Typically, term assurance is only on a conventional without-profits basis.

Whole life and endowment assurances are typically written on a unit-linked or with-profits basis, with new conventional without profits business being rare.

Annuities tend mainly to be written on a conventional without-profits basis, although are often also available on an index linked basis [and in some countries on a with-profits basis].

[2]

iii.

Administration Cost of the product:

- Unit linked products are likely to incur more administration cost compared to others as they
 - require creation and maintenance of units and detailed system set up
 - Would need more detailed and more frequent communication to policyholders than all other structures.
- Conventional without profit products are likely to cost least as system requirements and disclosures to policyholders are least.

- With profits structure would be in between, requiring segregation of funds, determination of equitable profit distribution and need to meet policyholders' reasonable expectations.

Riskiness of the product

- Conventional without profit structure carries most risk to the insurer as all the terms are guaranteed.
- Unit linked carry least risk, as the most significant investment risk can be fully transferred to policyholders.
- With profits structure might carry higher risk than unit linked structure, but significantly lower than the without profits structure. Smoothing mechanism and need to ensure policyholder reasonable expectation or treating customers fairly conditions might restrict the ability to pass on the adverse experience to policyholders.

Profitability

- Conventional without profit has relatively higher profit margins due to higher risk. However it may not be possible to sell significant volumes of these highly profitable products due to competition.
- Unit linked structure provides smaller per policy profits, but has greater potential to generate larger volumes and hence higher profit on portfolio level.
- With profits typically have very small profit margins since major proportion of the emerging surplus gets distributed to the policyholders.

[6]

[12 Marks]

Solution 3:**(i) Profit distribution and investment strategies:*****Additions to benefits method***

- The guarantees (the sum assured plus reversionary bonuses) typically are kept low in order to defer the distribution of profit. This deferral gives the company the freedom to invest in real assets such as equities and property.
- This leads to higher expected investment returns and hence higher expected maturity benefits.
- A strategy of higher guarantees, with correspondingly less real investment content, is also possible. This would be disclosed or evident in the sales literatures.
- Other sources of profit, such as from expenses and mortality, would be included in the surplus distributed to policyholders, thereby increasing expected returns.
- If traditional without profit policies are written with the support of the participating fund, the policyholders might also share in the profits from the company's without-profits business, again increasing the expected return.

- Broad equity of profit distribution compared with other cohorts/tranches of with-profits policyholders would be considered.
- Payouts could therefore be affected by cross-subsidy either way.
- Payouts could be considerably smoothed over time, hence individual years' payouts could significantly vary compared with asset shares at maturity.
- Some contribution might be made from the policy's profits to the company's estate (or *vice versa*) over the duration of the policy, decreasing (or increasing) the likely level of policy payouts accordingly.

Revalorization method

- A guaranteed sum assured is the minimum level of payout under the contract.
- Some (high) fixed percentage of each year's investment profit will be added to the reserves.
- Usually, both benefits and premiums are increased to the same extent.
- Alternatively the benefit only will be increased, but by a lower proportion.
- No other sources of profit would usually be distributed.
- Because there is very little profit deferral, the company would probably need to invest in matching assets to protect solvency, which would mean investing mostly in fixed-interest stocks of appropriate term.
- However the distribution of the investment profit is likely to be very equitable between policyholders (as it is being distributed proportionately to reserves)
- It is likely that the revalorization method will produce lower expected returns than the additions to benefits approach, in relation to premiums paid:
 - investments are likely to be weighted more towards fixed interest rather than real assets
 - expense and mortality profits are not distributed
 - profits from other business are not distributed.

[9]

(ii) This can arise if, under the additions to benefits method:

- the returns from equities and properties have been worse than fixed interest leading to overall lower return than under the revalorization method
- net losses have been incurred from expenses and mortality
- policyholders have shared in losses that have been incurred from the non-profit business written in the with- profits fund
- a particular policy class or tranche was receiving lower than its fair share of the actual contribution to profit (under the "broad equity" approach)
- payouts were lower than asset shares due to smoothing
- the company adopted a strategy of making a net contribution to the estate over the duration of the policy.

[3]

[12 Marks]

Solution 4:**(i)**

- The assumption should reflect the expected future experience of the lives who will take out the contract being priced.
- The company does not have any experience in writing group insurance business, and therefore would not have a group mortality basis to use as a starting point.
- However, since it has been selling term insurance business in the market, the mortality rates used for this individual business could be used as a starting point...
- ...with modifications made to reflect the differences expected in experience since this is a group portfolio:
 - Mortality would be expected to be lighter than the individual business since only lives that are currently in full time employment are covered, and lives that are too sick to work get automatically excluded.
 - In addition, there is lesser chance of anti-selection since the decision to buy the insurance (and the choice of the sum assured) does not rest on the life to be insured, but instead on the employer, with the sum assured usually determined using a formula dependent on the salary of the life insured.
 - However, this might also mean that the life insured is not necessarily financially aware, and might have worse mortality than in the case of the financially aware, and hence, arguably, better class of lives under individual business.
 - The terms might be different based on the occupation class of the employee, while the mortality rates under the individual insurance would assume a mix of occupation class based on the general profile of the insured population.
- In addition, advice would be sought from the reinsurer to determine the mortality basis. Given that this market is large, the reinsurer is likely to have experience and be able to provide valuable advice.
- Given that this is a large market, industry tables might exist specifically for group insurance, or else the differential for various occupation classes generally used in the market may be available.
- Due to the large market size, for the larger schemes past historical experience data might be available, allowing the mortality assumption be set for each scheme based on this past data...
- ...modified for any changes expected in future, for example due to change in management / hiring practices of the employer or the age/ gender / profile mix of the employees.
- Usually a single rate (or a rate varying by occupation classes) would be quoted irrespective of the age / gender. Therefore, it would be important to understand the mix of business expected and reflect this in the mortality rates before proving the quotation.

[5]

(ii)

- To limit the amount paid on a claim from the covered schemes
- To limit the total claims payout, particularly important for schemes with large members
- To reduce the claims payout fluctuations
- To receive technical assistance, especially since group life is new to the insurer
- To reduce the new business strain
- Increase profits, return or risk-adjusted return on capital (by reducing risk cost or increasing volumes)
- Reduce overall capital requirements by using a reinsurer's capital. Reinsurers may have lower capital requirements due to risk diversification or regulatory position.
- Allow aggregation of risks that the cedant cannot manage on its own, so allowing manufacture of product lines.

[2]

(iii)

- **Quote share**
 - + The reinsurer would cover a proportion of risk on each life, and would therefore provide a set of reinsurance rates that would factor in the risks from all the lives. This would be beneficial while pricing since the insurer is new in this area of business.
 - + The reinsurer has a vested interest in each life, and therefore may provide greater underwriting, claims and administrative assistance.
 - The reinsurer may impose restrictions on the size of the group / occupations that may be reinsured etc, restricting the marketing freedom
 - Each life would be reinsured even if it falls within the risk appetite of ABC. This is likely to lead to reduced profits since the reinsurance rates are also likely to have a profit margin in them.
 - The reinsurer may wish to charge a different premium for each group based on its past experience. This might require a referral to them for each group, delaying the response time in sending quotations to the employer
- **Individual Surplus**
 - + Only the lives that are beyond the insurer's risk appetite need to be reinsured. Therefore profits can be retained on the lives that are within the insurer's risk appetite.
 - The reinsurer rates provided to the insurer would only reflect the risk on the lives that are to be reinsured (i.e. higher ticket sizes). Therefore, it may not be possible to rely entirely on the reinsurance pricing to derive the direct insurance rate.
 - The reinsurer may impose restrictions on high ticket sizes that may be written overall. Thus restricting the marketing freedom.

- It might be possible to negotiate, as a part of the treaty, that if the reinsurer does not wish to cover a particular group, then this group could be offered to other reinsurers in the market on a facultative basis. However, this would also increase the turnaround times.
- + For groups that are entirely retained, there would be greater pricing freedom since the reinsurance rates would not need to be factored in.
- + For groups that are entirely retained, no referral to the reinsurance company might be required, leading to potentially faster turnaround times.
- For groups / lives that are entirely retained, the reinsurer may not provide support for underwriting / claims processing.
- Excess of loss reinsurance
 - Could be used to insure against aggregation of risk
.....Since this is group insurance, there might be employer / employee schemes such that many of insured's may be operating from the same office locations. In such cases the aggregation risk would be high.
 - + The cover may be difficult to obtain, or may be expensive in case the reinsurers do not have experience in this market.
 - + Renegotiation may be required every year.
 - Since the cover is on an aggregate basis, the reinsurer would have no incentive to provide support in underwriting / claims processing for individual members or groups. This might be valuable for the company since it is entering a new segment.
 - Similarly, there would be no way to use the reinsurance rates to derive the direct insurance rates.
 - The reinsurer may impose restrictions on the overall size of a group that may be reinsured to manage the aggregation risk faced by them. This might restrict marketing freedom.
 - + Turnaround times for individual groups may be faster since none of the decisions would require a referral to the reinsurer.

Options: Catastrophe reinsurance or Stop Loss reinsurance

- Catastrophe reinsurance would allow protection against a single event making is useful for terrorist activities / natural or man-made disasters etc.
- Stop Loss reinsurance would allow protection against the total losses in a given time period, usually a year. This might be more useful for protection in areas where a risk exists of smaller scale disasters that could occur frequently.

[10]

(iv)

Assumption:

- Ignoring IBNR.

The data shows the following:

Year	Average members during the year	No. of deaths during the year	Crude mortality rate per 1000 (Deaths / avg. members)
201X	200,246	213	1.0637
201X-1	187,974	169	0.8991
201X-2	199,783	201	1.0061
201X-3	156,846	172	1.0966
201X to 201X-3	744,849	755	1.0136

Observations:

- The crude mortality rate in the year 201X-1 seems materially lower than the crude rate in other years.
- The crude mortality rate for the year 201X appears to significantly higher than the mortality rates for the years before that.
- This suggests that the experience data for the year 201X-1 and 201X might need to be investigated further.
- It might be possible that the deaths have been assigned to an incorrect year while recording causing the skewness
- Or this might only be statistical variance.

Additional data:

- Movement data for the members along with the date the claim was reported – this might help check whether the claims are being assigned to the correct year on a claims occurring basis, instead of a claims reported basis.
- Age / Gender splits for each of the years to see if any demographic changes have led to the variance in the death rates.
- Any epidemic, etc. which led to a high death rate in Year 201X
- Reasons for exits, particularly in year 201X-1, where the number of members drop considerably.

[6]

[23 Marks]

Solution 5:

- (i) Embedded value is the present value of future shareholder profits in respect of the existing business of a company, including the release of shareholder-owned net assets.

It can be calculated as the sum of:

- The shareholder-owned share of net assets
where net assets are defined as the excess of assets held over those required to meet liabilities, and
- The present value of future shareholder profits arising on existing business.

[2]

- (ii) This decision would impact the general business environment for the company.

In particular:

- The marketing department may be right, since publishing the Embedded Value may help shareholders better understand the intrinsic value of the business written to date. If the results demonstrate stability, they may help build investor confidence.
- Unlikely to cause any change in the customer's propensity to purchase the products since it is the present value of the shareholder's profit, not policyholders.

Other factors to be considered would be similar to those considered while launching a new product:

- Competition: If competitors are already publishing this information, then the company will have to ensure that its embedded value results are competitive in order to attract investors.
If no other competitors are publishing this information then publishing might help create an image of transparency.
- Risk: It might expose the company to unnecessary reputational risk.
- Expenses: the embedded value may need to be reviewed more thoroughly, incurring expenses. A cost benefit analysis should be considered.
- Legislation: Rules and regulations might exist around publishing Embedded Values. These would need to be considered in depth.
- Level of disclosure to be made needs to be considered. Will assumptions be disclosed, for example?
- The impact of sensitivities disclosed on the investors and other stakeholders needs to be considered.
- Whether we would be able to consistently declare the embedded value going forward.
- The detail required to ensure that the investors understand the results.

[5]

(iii) a. Mortality:**I.**

- The pure term insurance product is likely to be most sensitive to change in mortality assumptions
...since it would have the highest sum at risk on death out of the three contracts.
- An increase in mortality would decrease the future expected profit from the pure term insurance product significantly, and therefore reduce the embedded value.

II.

- For the unit linked product, a higher mortality would not cause a huge negative cashflow immediately, since the sum assured is equal to the unit fund value.
- However, it would lead to a loss of future expected profitable cashflow from the policy. Therefore, it might cause a marginal decrease in the embedded value.

III.

- For the pure endowment contract, an increase in mortality might cause a slight increase in embedded value
... since it reduces the lives for which the survival benefit needs to be paid out.

b. Persistency:**I.**

- The pure term insurance will not have any surrender benefits. Increased persistency would increase the death benefits to be paid out
- However, increase in persistency will also increase the future expected premiums.
- Future increase in premiums is expected to offset the increase in death benefit and hence, increase the embedded value.

II

- The impact of increase in persistency would be opposite to mortality.
- Since the base persistency assumption would be significantly higher than the base mortality assumption, the impact is expected to be significantly higher in magnitude.

III

- During the earlier years, the surrender penalty of 30% may be insufficient to offset the loss of future profit from the product. Therefore, the increase in persistency would increase the future profits.
During the later years, the impact would be the opposite.
- The overall impact would depend on the vintage of the block of business i.e. whether more policies are in the initial years or later years.

c. Risk Discount Rate:**I.**

- Increasing the Risk discount rate would impact all the three products, assuming they are profitable, in a similar manner.

... it would reduce the embedded value since the future profits would be discounted at a higher rate.

[8]

[15 Marks]

Solution 6:

(i) Risks

Low unit growth –

- leading to insufficient management charges, to recoup the expenses incurred.
- On death the guarantee of $x\%$ of the single premium paid is more likely to bite, so the cost of providing death benefit may prove higher than expected.
- If unit growth has been poor or there is a sudden fall in unit price, then the benefit may be seen as poor value for money. This may cause bad publicity and affect new business levels.

Mortality

- The risk is that mortality will be higher than that assumed in the pricing basis. The risk is greatest for cases with $x = 300$.
- Lives choosing $x = 100$ are not medically underwritten, so there is a risk that mortality rates are worse for these lives than expected. On the other hand, people in very poor health are unlikely to take out this policy purely for the sake of the “free” cover, because they will only be getting back the same money that they paid in to the policy in the first place.

High expenses

- There is a risk that expense levels will be higher than expected, leading to reduced profits.
- This may be compounded by low new business volumes, causing fixed overheads to be spread over fewer policies than expected.
- Expense inflation may also be higher than allowed for in the pricing basis, leading to an erosion of profits.
- However, this may be offset to some extent by higher nominal investment returns and hence higher fund management charges.

Variability of charges

- Charges to cover mortality and expenses are variable in “certain circumstances”. This will allow some of the risk to be passed on to policyholders.
- Depending on the exact nature of these circumstances, there is a risk that the company cannot increase charges to reflect higher than expected costs.
- Even if charges were fully variable, the extent to which the company can increase charges is limited by PRE and large increases are likely to lead to higher selective withdrawals and lower new business volumes.
- There will be a delay between the increased costs and the implementation of increased charges, because of the time taken to recognize the deterioration in the experience.

Withdrawals [persistency]

- There is a risk of more withdrawals than expected at times when the surrender value is greater than the asset share. This may occur very early on in the contract.
- Higher withdrawals would lead to lower future charges and hence profits.

Low average premium size

- Both the 1% initial charge and the 1% management charge represent a cross-subsidy between large and small policies, so there is a risk that total charges make an insufficient contribution to per-policy costs if the average policy size is lower than assumed when pricing the contract.

High new business volume

- There may be an initial capital strain when this policy is sold, as reserves on day 1 are at least equal to the surrender value, which [in the absence of any stated surrender penalty] is likely to be higher than the day 1 asset share. The company has a risk to its supply of capital [and therefore its supervisory solvency] if much higher than expected volumes of sales occur.

[10]

(ii) Investment strategy

- Investment should maximise the overall return on the assets. In order to minimise risk, the company should select investments that are appropriate to the nature, term and currency of the liabilities. The company may depart from the above depending on the level of free assets.
- The unit liabilities should be matched by units in a separate investment fund.
- Usually there will be a range of funds the policyholder can choose from to suit his/her risk appetite – e.g. Balanced/managed; equity; bond; property; cash.
- The investments in each fund must be aligned with the investment objective and the asset mix of each fund as indicated in the communication material given to the policyholders.
- Some cash will be needed in each fund to cover claims without being a forced seller of investments.
- The funds should not gear their exposure to markets by borrowing cash or shorting stock.
- Non-unit liabilities need to be held in respect of expenses & mortality risk. In practice it will be difficult to match exactly, but could be matched by fixed interest/cash/index linked or real assets.
- The investment strategy would need to factor in that while the allocation charge is received at one go, future income stream in form of management charge and mortality charge will have to be invested monthly.
- Expenses may be matched by index linked bonds.
- Cash would be needed for liquidity purposes.

[5]

(iii)

- Higher death cover and related unit deductions will make the maturity guarantee onerous, if policyholders opt higher death cover option (higher x% - 300% eg.,) there is higher probability that the investment yield is not enough to meet the guaranteed accumulation
- This is worsened at the higher ages, where the mortality costs are higher
- Current charge structure would have been designed without having regard to the cost of guarantee.
- Hence the current charge structure may not meet profit criteria after allowing for the cost of guarantee.

[2]

(iv)

- The death cover may be restricted. For example, limiting x% to 200% instead of 300% of the single premium.
- Entry age might be restricted to age 40/45 which would limit the mortality charge to be deducted.
- Offering a new fund under the product which primarily invests in debt instruments and minimal participation in equities, with the guarantee applicable with this fund only.
- A guarantee charge may be separately charged
 - Considering the comparative products if available for the charging structure as % of fund or otherwise
 - Ensuring that the profit criteria are met.
- However, an additional charge in the form of guarantee charge will reduce the projected fund further and the guarantee may bite.
- Any of the above suggestions though will have an impact on the marketability of the product and the potential volumes.

[3]

[v] Determination of the guarantee cost

- Determine a range of model points based on expected mix of business using gender, age at entry, unit fund link and policy size
- Construct a single-policy cash-flow projection model that can be used for each model point.
- Project unit fund values for each year, including the guaranteed charging structure, using assumptions about unit growth rates.
- As base projection calculate net present values (NPVs) for each model point for the standard policy, **excluding** the maturity guarantee.
 - Initially need to project on the (prudent) supervisory reserving basis to establish the level of non-unit reserves required each year, if any.

- Then project the expected profit flows using the (realistic) pricing basis, allowing for the supervisory non-unit reserves, and discount at the risk discount rate to obtain the NPV.
- Next rerun the model including the guaranteed maturity pay out.
 - This will require a stochastic model of investment yields and inflation rates at the projected maturity date.
 - In those simulations that produce maturity value lower than guaranteed, the additional cost of providing the guaranteed maturity would be calculated.
- A large number of simulations would be run, and an average cost of the guarantee obtained from it.

Determination of suitability of charge towards the guarantee cost

- The unit fund and cash-flow projection models would then incorporate the additional cost of guarantee charge.
 - The model would again have to be run first on a prudent basis, so that the non-unit reserves can be established.
 - These reserves must enable the company to meet the estimated cost of the guarantee at maturity, based on a prudent assessment.
 - This cost would be found as the upper (say) 97.5th percentile of the distribution of simulated outcomes of the guarantee cost.
 - The new reserves are then incorporated into the premium basis cash flow model as before, now including a realistic value of the expected cost of the guarantee at maturity.
 - This realistic value is taken as the average of the simulated outcomes from the stochastic model.
- The NPV is then recalculated, and compared with that previously calculated (for the same model point) without the guarantee.
- Cases that show reduced NPVs would indicate undercharging for the guarantee.
- Comparing the results over the range of model points would indicate whether the additional charging rate ought to vary by entry age, premium size *etc*, or whether an alternative charging structure might be appropriate (*eg* have a monetary policy fee, or express as a proportion of single premium).
- All results should be sensitivity tested, using different assumptions, including different parameter values for the probability models.
- Competitors' charging rates for similar guarantees should be checked, if they exist. If the rates appear uncompetitive, choosing a different charging structure could avoid unfavourable comparisons.

[11]

[31 Marks]
